MATHS MANTHAN
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Subject: Mathematics | **WEEKLY CHAPTER TEST – 05 (POLYNOMIALS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 2: Polynomials** | **MM: 20**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, zero skipped steps, and final boxed answers with proper units.
2. In alternative (**OR**) sub-questions, complete step-by-step solutions and equal step marking apply to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Roots rejected by physical domain/restrictions are explicitly stated.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

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Q1. If α, β are the zeroes of the polynomial $ax^2 - 5x + c$ such that $\alpha + \beta = \alpha\beta = 10$, then: [1]

(A) $a = 5,\ c = \dfrac{1}{2}$ **(B)** $a = 1,\ c = \dfrac{5}{2}$

(C) $a = \dfrac{1}{2},\ c = 5$ **(D)** $a = \dfrac{1}{2},\ c = 10$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C)

Comparing $ax^2 - 5x + c$ with $Ax^2 + Bx + C$, we have $A = a$, $B = -5$, $C = c$. We know Vieta's formulas:

Formula Used: $\alpha + \beta = -\dfrac{B}{A}, \qquad \alpha\beta = \dfrac{C}{A}$

Applying $\alpha + \beta = 10$ and $\alpha\beta = 10$:

$$-\frac{(-5)}{a} = 10 \implies \frac{5}{a} = 10 \implies a = \frac{5}{10} = \boxed{\frac{1}{2}}, \qquad [0.5\text{M}]$$

$$\frac{c}{a} = 10 \implies c = 10a = 10\left(\frac{1}{2}\right) = \boxed{5}. \qquad [0.5\text{M}]$$

Q2. The zeroes of the quadratic polynomial $p(x) = 4x^2 - 12x + 9$ are: [1]

(A) $\dfrac{3}{2}, \dfrac{3}{2}$ (B) $-\dfrac{3}{2}, -\dfrac{3}{2}$ (C) $3, -3$ (D) $\dfrac{2}{3}, \dfrac{2}{3}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

We know the square identity $(A - B)^2 = A^2 - 2AB + B^2$:

Formula Used: $(2x - 3)^2 = 4x^2 - 12x + 9$

Setting $p(x) = 0$:

$$(2x - 3)^2 = 0 \implies 2x - 3 = 0 \text{ (repeated)} \implies x = \boxed{\frac{3}{2}, \frac{3}{2}}. \qquad [1\text{M}]$$

Q3. If α, β are the zeroes of $f(x) = x^2 - 5x + k$ such that $\alpha - \beta = 1$, then the value of k is: [1]

(A) 6 (B) 4 (C) 5 (D) 12

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A)

From $f(x) = x^2 - 5x + k$, we know Vieta's relations:

Formula Used: $\alpha + \beta = -\frac{b}{a} = 5, \qquad \alpha\beta = \frac{c}{a} = k$

Adding $\alpha + \beta = 5$ and $\alpha - \beta = 1$ gives $2\alpha = 6 \implies \alpha = 3$, so $\beta = 5 - 3 = 2$. Hence:

$$k = \alpha\beta = (3)(2) = \boxed{6}. \qquad [1M]$$

Q4. If the zeroes of $x^2 - kx + 6$ are in the ratio $2 : 3$, then the positive value of k is: [1]

(A) 5 (B) -5 (C) 6 (D) 1

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A)

Let the zeroes in ratio $2 : 3$ be $2m$ and $3m$ $(m \neq 0)$. We know Vieta's formulas:

Formula Used: $\alpha + \beta = -\frac{b}{a} = k, \qquad \alpha\beta = \frac{c}{a} = 6$

Thus $(2m)(3m) = 6 \implies 6m^2 = 6 \implies m^2 = 1 \implies m = \pm 1$. Taking the positive value $(m = 1)$:

$$k = 2m + 3m = 5m = 5(1) = \boxed{5}. \qquad [1M]$$

Q5. Assertion-Reason: [1]

Assertion (A): If one zero of $(k^2 + 16)x^2 + 13x + 8k$ is reciprocal of the other, then $k = 4$.

Reason (R): If zeroes of $ax^2 + bx + c$ are reciprocals of each other, then $a = c$.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A) Both A and R are true and R is the correct explanation of A.

We know that for $ax^2 + bx + c$ $(a \neq 0)$, if zeroes are α and $\frac{1}{\alpha}$:

Formula Used: $\alpha \cdot \frac{1}{\alpha} = \frac{c}{a} \implies \frac{c}{a} = 1 \implies a = c$

Applying $a = c$ to $(k^2 + 16)x^2 + 13x + 8k$ gives $k^2 + 16 = 8k \implies k^2 - 8k + 16 = 0 \implies (k - 4)^2 = 0 \implies k = 4$. Both **(A)** and **(R)** are true, and **(R)** explains **(A)**. [1M]

Q6. Form a quadratic polynomial whose zeroes are $3+\sqrt{2}$ and $3-\sqrt{2}$. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let $\alpha = 3+\sqrt{2}$ and $\beta = 3-\sqrt{2}$. We know the quadratic construction formula:

Formula Used: $p(x) = k\big[x^2 - (\alpha+\beta)x + \alpha\beta\big], \quad k \neq 0$

Computing the sum and product of zeroes:

$$S = \alpha + \beta = (3+\sqrt{2}) + (3-\sqrt{2}) = 6, \qquad \textbf{[1M]}$$
$$P = \alpha\beta = (3+\sqrt{2})(3-\sqrt{2}) = 3^2 - (\sqrt{2})^2 = 9 - 2 = 7. \qquad \textbf{[0.5M]}$$

Taking $k = 1$, the required quadratic polynomial is $\boxed{p(x) = x^2 - 6x + 7}$ (or $k(x^2 - 6x + 7)$). **[0.5M]**

Q7. If the squared difference of the zeroes of the quadratic polynomial $f(x) = x^2 + px + 45$ is equal to 144, find the value(s) of p. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Let α and β be the zeroes of $f(x) = x^2 + px + 45$ ($a = 1$, $b = p$, $c = 45$). We know Vieta's formulas and the algebraic identity:

Formula Used: $\alpha + \beta = -p, \quad \alpha\beta = 45, \quad (\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$

Using the given condition $(\alpha - \beta)^2 = 144$:

$$(\alpha+\beta)^2 - 4\alpha\beta = 144 \qquad \textbf{[1M]}$$
$$(-p)^2 - 4(45) = 144$$
$$p^2 - 180 = 144 \implies p^2 = 324 \qquad \textbf{[1M]}$$
$$\implies \boxed{p = \pm 18} \text{ (i.e., } p = 18 \text{ or } p = -18). \qquad \textbf{[1M]}$$

Q8. If α and β are the zeroes of the quadratic polynomial $f(x) = 2x^2 - 5x + 7$, find a quadratic polynomial whose zeroes are $2\alpha + 3\beta$ and $3\alpha + 2\beta$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Comparing $f(x) = 2x^2 - 5x + 7$ with $ax^2 + bx + c$, we have $\alpha + \beta = \frac{5}{2}$ and $\alpha\beta = \frac{7}{2}$. **[0.5M]**

We know the formula to construct a polynomial with new zeroes $\alpha' = 2\alpha + 3\beta$ and $\beta' = 3\alpha + 2\beta$:

Formula Used: $p(x) = k\big[x^2 - S'x + P'\big], \quad S' = \alpha' + \beta', \quad P' = \alpha'\beta'$

Calculating S' and P' systematically:

$$S' = (2\alpha + 3\beta) + (3\alpha + 2\beta) = 5(\alpha + \beta) = 5\left(\frac{5}{2}\right) = \frac{25}{2}, \qquad \textbf{[1M]}$$
$$P' = (2\alpha + 3\beta)(3\alpha + 2\beta) = 6\alpha^2 + 13\alpha\beta + 6\beta^2$$
$$= 6\big[(\alpha+\beta)^2 - 2\alpha\beta\big] + 13\alpha\beta = 6(\alpha+\beta)^2 + \alpha\beta$$
$$= 6\left(\frac{5}{2}\right)^2 + \frac{7}{2} = \frac{75}{2} + \frac{7}{2} = \frac{82}{2} = 41. \qquad \textbf{[1M]}$$

Therefore $p(x) = k\left(x^2 - \frac{25}{2}x + 41\right)$; setting $k = 2$ gives $\boxed{p(x) = 2x^2 - 25x + 82}$. **[0.5M]**

Q9. Find the zeroes of $f(x) = 4x^2 + 5\sqrt{2}\,x - 3$ and verify the relationship between the zeroes and the coefficients. **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Comparing $f(x) = 4x^2 + 5\sqrt{2}\,x - 3$ with $ax^2 + bx + c$, we have $a = 4$, $b = 5\sqrt{2}$, $c = -3$. We know:

Formula Used: $\alpha + \beta = -\dfrac{b}{a}, \qquad \alpha\beta = \dfrac{c}{a}$

Step 1 – Factorisation: Product $a \cdot c = -12$; splitting $5\sqrt{2} = 6\sqrt{2} - \sqrt{2}$ (since $(6\sqrt{2})(-\sqrt{2}) = -12$):

$$\begin{aligned} f(x) &= 4x^2 + 6\sqrt{2}\,x - \sqrt{2}\,x - 3 \\ &= 2\sqrt{2}\,x(\sqrt{2}\,x + 3) - 1(\sqrt{2}\,x + 3) = (2\sqrt{2}\,x - 1)(\sqrt{2}\,x + 3). \end{aligned} \qquad \textbf{[1M]}$$

Zeroes are $\alpha = \boxed{\dfrac{1}{2\sqrt{2}} = \dfrac{\sqrt{2}}{4}}$ and $\beta = \boxed{-\dfrac{3}{\sqrt{2}} = -\dfrac{3\sqrt{2}}{2}}$. **[1M]**

Step 2 – Verification:

$$\alpha + \beta = \frac{\sqrt{2}}{4} - \frac{3\sqrt{2}}{2} = \frac{\sqrt{2} - 6\sqrt{2}}{4} = -\frac{5\sqrt{2}}{4}, \qquad -\frac{b}{a} = -\frac{5\sqrt{2}}{4}, \qquad \textbf{[0.5M]}$$

$$\alpha\beta = \left(\frac{1}{2\sqrt{2}}\right)\left(-\frac{3}{\sqrt{2}}\right) = -\frac{3}{4}, \qquad \frac{c}{a} = \frac{-3}{4}. \qquad \textbf{[0.5M]}$$

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Q10. Case Study: Flight of an Incline Football Kick

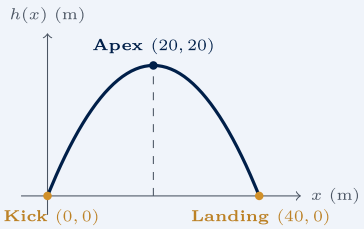
A football's vertical flight is modelled by $h(x) = -\frac{1}{20}x^2 + 2x$, where x is horizontal distance travelled (m) and $h(x)$ is height above the turf (m). [4]

(i) State the geometric shape traced by the football and determine whether the parabola opens upward or downward. **[1]**

(ii) Find the horizontal distance x where the football touches the turf again. **[1]**

(iii) **(a)** Determine the maximum height attained by the football during its flight. **OR (b)** The goalkeeper stands 30 m from the kick. If the crossbar is 2.44 m high, determine whether the ball lands inside the goal or flies over the crossbar. **[2]**

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]



Trajectory Overview: Kicked from origin $(0, 0)$, the ball rises symmetrically to its apex $(20, 20)$ at mid-range and lands on the turf at $(40, 0)$.

Formula Used: $h(x) = -\frac{1}{20}x^2 + 2x = -\frac{1}{20}(x-20)^2 + 20$

(i) Curve shape & direction: Since $h(x)$ is quadratic with $a = -\frac{1}{20} < 0$, the trajectory is a $\boxed{\text{parabola opening downward}}$. [1M]

(ii) Horizontal landing distance: Setting $h(x) = 0$:

$$-\frac{1}{20}x^2 + 2x = 0 \implies x\left(-\frac{x}{20} + 2\right) = 0 \implies x = 0 \text{ (kick) or } \frac{x}{20} = 2 \implies \boxed{x = 40 \text{ m}}. \quad [1M]$$

(iii)(a) Maximum height: Completing the square in $h(x)$:

$$h(x) = -\frac{1}{20}(x^2 - 40x) = -\frac{1}{20}[(x-20)^2 - 400] = -\frac{1}{20}(x-20)^2 + 20. \quad [1M]$$

Since $-\frac{1}{20}(x-20)^2 \le 0$, maximum height occurs at $x = 20$ m and equals $\boxed{20 \text{ m}}$. [1M]

OR (Alternative Part (b) – Crossbar clearance at $x = 30$ m):

$$h(30) = -\frac{1}{20}(30)^2 + 2(30) = -\frac{900}{20} + 60 = -45 + 60 = \boxed{15 \text{ m}}. \quad [1M]$$

Comparing with crossbar height 2.44 m (15 m > 2.44 m):

$\boxed{\text{The ball flies well OVER the crossbar.}}$ [1M]

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